

Where go the water used for cooling data center ?

Posted by u/AFuckingBadPlayer • 0 points • 11 comments

Ok so, data center use a LOT of water to cool down but where does it go ?
Does it become contaminated in any way ?
Because if they pump at one point why can't they injected it back ? Is it because it has become hotter ?

Thanks you in advance

Comments

u/rwiddi72 • 2 points • 2025-10-26 20:06 UTC

There's a place nearish me that uses the water to heat a swimming pool

u/anti-beep • 3 points • 2025-10-26 19:41 UTC

Depends on the data center. Sometimes it's in a closed or semi-closed loop, then it mostly stays in the cooling system.

But when you need *really* efficient cooling, you use evaporative cooling. The water still isn't being 'consumed', but you're dispersing it into the atmosphere.

This messes with the natural water cycle, which can destroy local ecosystems or cause scarcity.

u/AFuckingBadPlayer • 1 points • 2025-10-26 19:45 UTC

Okay thank you for your answer.

u/kattxni • 2 points • 2025-10-26 19:41 UTC

Water used for cooling usually ends up in local water systems. It can get warmer but not always contaminated. Some places reuse it after cooling but others strip it away. It depends on local regulations and tech. So injecting it back might not be allowed in some places. Just another layer to the complex system

u/East-Bike4808 • 2 points • 2025-10-26 19:36 UTC

At ours we circulate it. Return water goes to a cooling tower and gets cold again. There's a LOT of water moving around but for the most part it's just moving in a loop and not going somewhere afterwards. It gets topped off continually because of water evaporating (in the cooling towers).

u/AFuckingBadPlayer • 1 points • 2025-10-26 19:39 UTC

Okay thanks for your answer. So when I see news about the data center using all the water in an area, so much that even farmers can't water their crops. Is it because of the amount of data centers or is it an extrapolation from the lack of water?

u/AgentElman • 2 points • 2025-10-26 19:46 UTC

The farmers are the ones using the water, not the datacenter.

Farming is 80% of the water usage in the U.S. And the farmers have special rights to water which causes them to not try to conserve it - so they are massively wasteful in their water usage.

Datacenters are less than 1% of water usage in the U.S.

u/AFuckingBadPlayer • 1 points • 2025-10-26 19:47 UTC

Ooooh then the french news are just spouting nonsense again !

u/East-Bike4808 • 2 points • 2025-10-26 19:42 UTC

That's mostly evaporated water lost due to evaporative cooling (how most places cool). It can be an issue if it's a very large datacenter in a water-scarce location to begin with.

u/quicksopemcjerhoff • 2 points • 2025-10-26 19:32 UTC

I'm curious about this as well. I can't imagine them running tap water constantly to cool computers when they could just make a closed loop system with giant radiators.

u/AFuckingBadPlayer • 1 points • 2025-10-26 19:36 UTC

I guess they won't use tap water because tap water needs treatment and that would be a waste but i guess they pump it from somewhere ? They probably can pump it back from where it comes from. I know they do that for the cooling loop used in nuclear power plants (not all of them but some) here in France.